Alexandre Perozim de Faveri

Contact Information

EPFL SB MATH
CM 1 617 (Centre Midi)
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Research Interests

Analytic number theory and its interactions with harmonic analysis, automorphic forms, dynamical systems, spectral theory, and additive combinatorics.

Employment

École Polytechnique Fédérale de Lausanne (EPFL) 2025 – Present
Bernoulli Instructor

Stanford University 2023 – 2025
Postdoctoral Scholar

Institute for Advanced Study (IAS) 2022 – 2023
Member

Education

California Institute of Technology 2018 – 2022
Ph.D. in Mathematics
Advisor: Maksym Radziwiłł

Princeton University 2014 – 2018
A.B. in Mathematics
Advisor: Peter Sarnak

Preprints and Publications

10. **Optimal large sieve for fixed order characters**
Submitted (2026), 27 pp.
9. **The sixth moment of the Riemann zeta function**
Joint with M. Pandey
Submitted (2026), 19 pp.
8. **Non-orthogonality of the cubic and quartic large sieves via Rankin–Selberg**
Joint with A. Dunn and J. Hoffstein
Submitted (2026), 41 pp.
7. **Non-vanishing for quartic Hecke L-functions and ranks of elliptic curves**
Joint with C. Castillo and A. Dunn
Submitted (2026), 65 pp.
6. **Mass equidistribution for lifts on hyperbolic 4-manifolds**
Joint with Z. Shem-Tov
Submitted (2026), 31 pp.
5. **Non-vanishing for cubic Hecke L-functions**
Joint with C. David, A. Dunn, and J. Stucky
Submitted (2024), 93 pp.
4. **Non-escape of mass for arithmetic quantum limits on hyperbolic 4-manifolds**
Joint with Z. Shem-Tov
Submitted (2024), 29 pp.

3. **Simple zeros of** GL(2) L**-functions**
Journal of the European Mathematical Society (JEMS) 27 (2025), no. 7, 2985–3021

2. **The variance of closed geodesics in balls and annuli on the modular surface**
Advances in Mathematics 403 (2022), Paper No. 108390, 45 pp.

1. **Möbius disjointness for $C^{1+\varepsilon}$ skew products**
International Mathematics Research Notices. IMRN (2022), no. 4, 2513–2531

Awards and Fellowships

1. Scott Russell Johnson Dissertation Prize (Caltech Math Department) 2022
for the best graduate dissertation in mathematics
2. Scott Russell Johnson First-Year Prize (Caltech Math Department) 2019
for excellence in first-year graduate studies
3. Phi Beta Kappa 2018
4. Andrew H. Brown Prize (Princeton Math Department) 2017
for outstanding juniors in mathematics
5. Shapiro Prize for Academic Excellence (Princeton University) 2016
6. Fellow of Fundação Estudar 2014
7. IMO Bronze Medal 2014

Teaching

Instructor at EPFL

• Math 437	Calculus of Variations	Spring 2027
• Math 202(c)	Analysis III (Vector Calculus for Engineering)	Autumn 2026
• Math 327	Topics in Complex Analysis	Autumn 2025

Instructor at Stanford

• Math 20	Calculus (Head Instructor)	Spring 2025
• Math 152	Elementary Theory of Numbers	Winter 2024
• Math 110	Number Theory for Cryptography	Autumn 2023

Teaching Assistant at Caltech

• Math 3	Probability and Statistics	Winter 2022
• Math 17	Putnam Seminar (Instructor)	Fall 2021
• Math 11	Mathematical Writing	Spring 2021
• Math 17	Putnam Seminar (Instructor)	Fall 2020
• Math 3	Probability and Statistics	Winter 2020
• Math 110	Graduate Real Analysis	Fall 2019
• Math 3	Probability and Statistics	Winter 2019
• Math 2	Differential Equations	Fall 2018

Teaching Assistant/Grader at Princeton

• MAT 218	Accelerated Honors Analysis	Spring 2018
• MAT 325	Complex Analysis	Fall 2017
• COS 423	Theory of Algorithms	Spring 2017
• COS 226	Algorithms and Data Structures	Fall 2016
• COS 226	Algorithms and Data Structures	Spring 2016
• COS 226	Algorithms and Data Structures	Fall 2015

Service

Students Supervised

- ❑ Jan Michal Radominski-Lasek (Bachelor's thesis, École polytechnique) 2026

Seminars Organized

- ❑ Organizer of the Stanford Analytic Number Theory Seminar 2023 – 2025
- ❑ Organizer of the Caltech Graduate Student Seminar 2019 – 2020

Journal Refereeing

- ❑ Quick opinions: Annals of Mathematics • Journal of the American Mathematical Society • Journal d'Analyse Mathématique
- ❑ Referee: Forum of Mathematics. Pi • Communications in Mathematical Physics • International Mathematics Research Notices. IMRN • Journal of the London Mathematical Society • Mathematische Zeitschrift • Journal de Théorie des Nombres de Bordeaux • Proceedings of the American Mathematical Society • Acta Arithmetica • Journal of Number Theory • Forum Mathematicum • Canadian Journal of Mathematics • Bulletin of the Brazilian Mathematical Society • Bulletin of the Australian Mathematical Society • Canadian Mathematical Bulletin
- ❑ Reviewer: Mathematical Reviews

Other

- ❑ Lecturer for the Summer School on Analytic Number Theory 2025 (Xi'an, China)
- ❑ External expert for the Master's thesis of Erasmo Santini (EPFL, 2026)
- ❑ Editor of the AIM Problem List on Sarnak's Conjecture
- ❑ Problem Coordinator for IMO 2017 (Rio de Janeiro, Brazil)
- ❑ Grader for CHMMC and PUMaC

Research Visits

1. Centre de Recherches Mathématiques (CRM), April to July 2026
2. Sorbonne Université, April 2026
3. Georgia Tech, November 2024
4. Concordia University, October 2024
5. IAS, October 2024
6. UT Austin, April 2022
7. University of Bristol, November 2021
8. UT Austin, October 2021

Invited Talks

1. *Kansas State Number Theory Seminar*
   Kansas State University (online), October 2026
2. *The Fourth JNT Biennial Conference*
   Cetraro, July 2026
3. *Probability in Number Theory Seminar*
   Université de Montréal, June 2026
4. *Recent Perspectives on Moments of L-Functions*
   UT Dallas, June 2026
5. *ETIENE Joint Meeting, Bordeaux*
   Université de Bordeaux, April 2026
6. *University of Virginia Number Theory Seminar*
   University of Virginia (online), November 2025
7. *AMS Spring Eastern Sectional Meeting*
   Hartford, April 2025

8. *CRG Weekly Seminar Series*
   Online, January 2025
9. *Copenhagen Number Theory Seminar*
   University of Copenhagen (online), December 2024
10. *CDSNS Colloquium*
    Georgia Tech, November 2024
11. *Georgia Tech Number Theory Seminar*
    Georgia Tech, November 2024
12. *UC Davis Algebraic Geometry and Number Theory Seminar*
    UC Davis, November 2024
13. *Quebec-Vermont Number Theory Seminar*
    Concordia University, October 2024
14. *Southern California Number Theory Day*
    UC Irvine, October 2024
15. *Purdue Analytic Number Theory and Harmonic Analysis Seminar*
    Purdue University, October 2024
16. *Joint IAS/PU Groups and Dynamics Seminar*
    Institute for Advanced Study, October 2024
17. *Shanks Conference Series*
    Vanderbilt University, May 2024
18. *Stanford Number Theory Seminar*
    Stanford University, April 2024
19. *FRG Grad Seminar*
    Online, February 2024
20. *Around Frobenius distributions and related topics IV*
    Online, October 2023
21. *IAS Members' Colloquium*
    Institute for Advanced Study, April 2023
22. *Crosscurrents in number theory, analysis and geometry*
    Université Sorbonne Paris Nord (online), December 2022
23. *UCLA Number Theory Seminar*
    University of California, Los Angeles, April 2022
24. *First International Workshop in ANT at UNIST*
    Ulsan National Institute of Science and Technology (online), April 2022
25. *Heilbronn Number Theory Seminar*
    University of Bristol, November 2021
26. *Number Theory Seminar*
    University of Mississippi (online), October 2021

PERSONAL Citizenship: Brazilian